Local Burden and Demographic Disparities in Alzheimer's Disease and Related Dementias: A Generalizability Analysis of the Stanford ADRC

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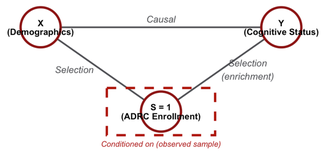
Background

- **Alzheimer's disease (AD) and AD-related dementias (ADRD)** affect over 6 million Americans, but local estimates of disease burden – particularly for diverse communities – remain limited.
- **National prevalence models often lack estimates for Asian Americans**, a critical gap for the Bay Area where Asian Americans represent a substantial share of the older adult population.
- Alzheimer's Disease Research Centers provide rich clinical data, but enroll a higher proportion of people with cognitive impairment than the general population. This **clinical enrichment** complicates generalization of ADRC findings.



Research Questions

1. What is the estimated local prevalence of AD/ADRD among adults aged 65+ in the ADRC's three-county catchment area?
2. Are there disparities in cognitive outcomes across racial/ethnic groups after demographic standardization?
3. How does clinical enrichment shape the ADRC sample relative to the target population?

Data & Methods

- **Data:** Stanford ADRC (529 adults 65+, NACC UDS v3, 2015–2024); California BRFSS (2,150 adults 65+ in Alameda, San Mateo, Santa Clara counties, 2015–2024); published coefficients from Dhana et al. (2023) and Matthews et al. (2019).
- **Q1 – Local AD Prevalence (Hybrid Model):** Dhana et al. published logistic regression coefficients for AD prevalence with age, sex, education, and race/ethnicity as predictors, but their model lacks an Asian American category. We constructed a hybrid model:

$$\text{logit}(P(\text{AD}_i)) = \hat{\alpha}^D + \hat{\beta}^D_{\text{age}} x_{\text{age}} + \hat{\beta}^D_{\text{sex}} x_{\text{sex}} + \hat{\beta}^D_{\text{edu}} x_{\text{edu}} + \hat{\beta}^A_{\text{race}} x_{\text{race}}$$

where D denotes published CHAP coefficients and $\hat{\beta}^A_{\text{race}}$ are log-odds ratios for AD by race/ethnicity (vs. White NH) estimated from the Stanford ADRC. We verified transportability of these ORs using case-control principles by testing for variation in clinical oversampling across groups. The model was applied to individual-level BRFSS data with survey weights. For ADRD, we used Matthews et al. (2019) rates via indirect standardization.

- **Q2 – Disparities (AIPSW):** We estimated prevalence ratios using augmented inverse probability of selection weighting:

$$\hat{\theta}_{DR} = \frac{\sum_{i \in T} \hat{\mu}(X_i) \cdot w_i^T}{\sum_{i \in T} w_i^T} + \frac{\sum_{i \in S} (Y_i - \hat{\mu}(X_i)) \cdot w_i^S}{\sum_{i \in S} w_i^S}$$

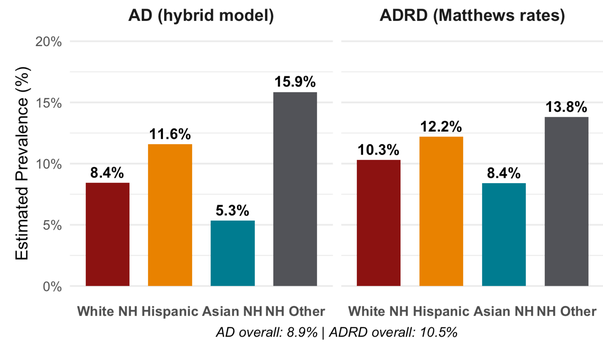
where T = BRFSS target population, S = ADRC source sample, $\hat{\mu}(X)$ = outcome model predictions, w^T = survey weights, and w^S = stabilized inverse odds of selection weights. This doubly robust estimator is consistent if either the participation model or the outcome model is correctly specified (Dahabreh et al. 2019). Variance was estimated via stratified bootstrap (1,000 replicates $\times$ 40 multiply imputed datasets with Rubin's rules).

- **Q3 – Enrichment:** We characterized ADRC referral pathways and compared ADRC-derived odds ratios to published population-based estimates.

Results

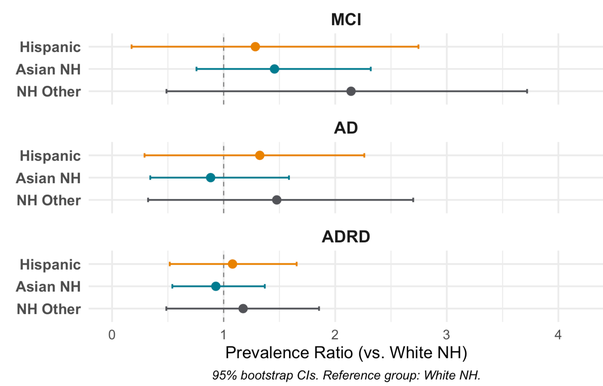
Local AD/ADRD Prevalence

Figure 1. Estimated AD/ADRD prevalence by race/ethnicity



Demographic Disparities

Figure 2. AIPSW prevalence ratios by race/ethnicity



Clinical Enrichment

Figure 3. Cognitive impairment by referral pathway

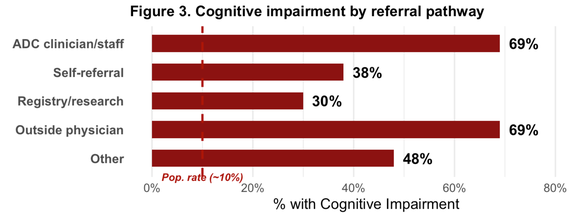
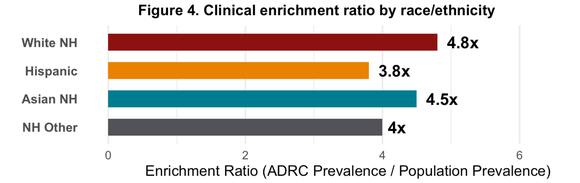


Figure 4. Clinical enrichment ratio by race/ethnicity



Conclusions & Future Directions

- Estimated AD prevalence is **8.9%** among adults 65+ in the three-county catchment, highest among Hispanic older adults (11.6%) and lowest among Asian Americans (5.3%).
- Clinical enrichment is substantial across all referral pathways. Demographic reweighting of ADRC data **does not enable estimation of population prevalence**, but under approximately constant enrichment across groups, it can support estimation of **racial and ethnic disparities** in cognitive outcomes.
- **Black NH adults are underrepresented** in the ADRC (n=12 at 65+) relative to the catchment population. Expanding recruitment is a priority.
- **Future directions:** Expand recruitment of underrepresented groups; extend to longitudinal trajectories and incident dementia; disaggregate the Asian NH category given known heterogeneity across subgroups (Mayeda et al. 2016).

Limitations

- The hybrid model assumes clinical enrichment is constant across racial/ethnic groups. This was tested empirically (ratios ranged from 3.8x to 4.8x) but with limited precision given subgroup sample sizes.
- Hispanic (n=58) and NH Other (n=18) ADRC subgroups yield wide confidence intervals for race-specific estimates.
- The ADRC's etiological composition (~45% Lewy body disease among ADRD cases) reflects Stanford's research focus and may not generalize to population-based etiological distributions.
- Black NH participants were collapsed into the NH Other category due to small sample size (n=12), precluding separate estimation of Black-White disparities.
- BRFSS captures only non-institutionalized adults, excluding nursing home residents among whom ADRD prevalence is substantially higher.
- The Dhana et al. CHAP coefficients were estimated in a Chicago population; age, sex, and education associations may not transfer perfectly to the Bay Area.
- Several covariates used in prior generalizability analyses (income, self-rated health, exercise, functional limitations) were unavailable in the NACC UDS and could not be included in the participation model.
- Hypertension was structurally missing in BRFSS even-numbered survey years (2016, 2018, 2020, 2022), addressed via multiple imputation but representing a limitation.

References

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Acknowledgments

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